Supra-Hodge Laplacians for Multi-View Reasoning:
A Higher-Order Spectral Framework for Coupled Simplicial Systems and Conflict-Aware Large Language Model Decision-Making

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Abstract

We introduce the *Supra-Hodge Laplacian*, a higher-order spectral operator defined on a family of coupled simplicial complexes. The construction unifies three mathematical ingredients: classical graph Laplacians, supra-Laplacians for multilayer systems, and Hodge Laplacians on simplicial complexes. Unlike graph-based operators that encode only pairwise relations, the proposed framework captures both intra-layer higher-order topology and cross-layer consistency through a block-structured operator acting on aligned chain spaces. Its associated quadratic energy functional decomposes into within-layer smoothness and curl penalties together with cross-layer disagreement terms, yielding a rigorous variational description of coupled higher-order relational systems.

We develop the theory in a math-first manner. After defining the operator formally, we establish its basic structural properties, including symmetry, positive semidefiniteness, and reduction to classical graph and supra-Laplacian settings in appropriate limits. We then interpret its spectrum: low-energy eigenspaces describe jointly smooth, cross-view coherent configurations; Fiedler-type modes identify conflict frontiers; and the higher-order components expose cycle-level inconsistencies that are invisible to purely pairwise methods.

As an application, we map multi-view large language model (LLM) reasoning problems into coupled simplicial systems whose layers encode semantic similarity, evidential support, task dependency, state-transition feasibility, execution order, and agent–tool interaction. In this setting, the Supra-Hodge Laplacian becomes a practical diagnostic and compression mechanism for reasoning. Proof-of-concept experiments on a synthetic 8-node scientific-planning case study and on 50 synthetic multi-view instances show that the proposed framework detects higher-order conflicts more accurately than pairwise baselines and preserves structurally relevant context during spectral compression. The central contribution of this work is therefore twofold: a new mathematical operator for coupled higher-order systems, and a concrete demonstration that this operator is useful for modern reasoning architectures.

1 Introduction

Many systems of interest are neither purely pairwise nor governed by a single relation type. Scientific workflows, planning systems, retrieval pipelines, biological interaction systems, and multi-agent decision processes all exhibit *heterogeneous* relational structure: the same set of entities may be

simultaneously linked by semantic affinity, causal dependence, evidential reinforcement, temporal order, feasibility constraints, and operational interactions. A graph captures only one relation type at a time, and a classical graph Laplacian inherits that limitation.

This paper develops a higher-order spectral operator for precisely this setting. The object of study is a *family of simplicial complexes* built over a shared entity universe, each complex encoding one structural view of the same system. The central question is then the following:

> Can one construct a single operator whose energy penalizes not only within-view roughness and higher-order inconsistency, but also disagreement across views?

The answer proposed here is the *Supra-Hodge Laplacian*. It extends the ordinary Laplacian in two orthogonal directions. First, it replaces graphs by simplicial complexes, thereby capturing higher-order interactions such as triangles, loops, and multi-way compatibility constraints. Second, it couples multiple complexes by a supra-operator that penalizes disagreement across layers. The resulting object is not merely a bookkeeping device: it has a natural quadratic energy, an interpretable spectrum, and useful reduction properties.

Although the operator is mathematically motivated, an important application comes from large language model (LLM) reasoning. Realistic reasoning traces are multi-view by nature. A plan may be semantically coherent but evidentially unsupported; it may satisfy task dependencies yet violate state-transition feasibility; a retrieved subgraph may look locally consistent while hiding a global cycle-level contradiction. These are precisely the kinds of failures for which pairwise graph methods are too weak. The mathematical framework developed here therefore provides a principled foundation for conflict-aware reasoning diagnostics.

The main contributions of this paper are:

1. We formalize the Supra-Hodge Laplacian as a block-structured operator on coupled simplicial chain spaces.
2. We derive its associated energy functional and establish its basic properties, including symmetry, positive semidefiniteness, and reduction to known operators.
3. We interpret its spectrum in terms of higher-order consistency, cross-view agreement, conflict frontiers, and low-energy structure.
4. We show how the operator can be instantiated in multi-view LLM reasoning, where it serves as a diagnostic and compression mechanism rather than a hard controller.
5. We provide proof-of-concept empirical evidence that the higher-order construction detects conflicts missed by pairwise baselines.

The paper is organized as follows. Section 2 reviews the necessary background from spectral graph theory, multilayer systems, and simplicial Hodge theory. Section 3 formally defines the Supra-Hodge Laplacian. Section 4 interprets its spectrum and energy. Section 5 discusses incremental and dynamical updates. Section 6 presents the application to LLM reasoning. Section 7 reports proof-of-concept experiments, and Section 9 discusses limitations and extensions.

2 Preliminaries

2.1 Classical Spectral Graph Theory

Let $G = (V, E, w)$ be a finite undirected weighted graph with $|V| = n$. Its adjacency matrix is $A \in \mathbb{R}^{n\times n}$ with entries $A_{uv} = w(u, v) \geq 0$, and its degree matrix is $D = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_u = \sum_v A_{uv}$. The combinatorial graph Laplacian is

$$L = D - A.$$

The Laplacian is symmetric positive semidefinite. For any $x \in \mathbb{R}^n$,

$$x^\top Lx = \frac{1}{2} \sum_{u,v\in V} A_{uv}(x_u - x_v)^2, \tag{1}$$

so L measures lack of smoothness across adjacent vertices. Its eigenvalues satisfy

$$0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n,$$

and the second eigenvalue λ_2 together with its eigenvector (the Fiedler vector) encodes the weakest nontrivial cut and underlies spectral clustering [1].

2.2 Multilayer and Supra-Laplacian Systems

Suppose now that the same entity set V carries k distinct relation types. A multilayer graph system may be written

$$\mathcal{G} = \{G^{(1)}, \ldots, G^{(k)}\},$$

with one graph per layer. If $L^{(i)}$ is the Laplacian of layer i and $\omega_{ij} \geq 0$ are inter-layer coupling weights, then a standard supra-Laplacian has block form

$$L_{\mathrm{sup}} = \begin{bmatrix} L^{(1)} + \sum_{j\neq 1} \omega_{1j} I & -\omega_{12} I & \cdots & -\omega_{1k} I \\ -\omega_{21} I & L^{(2)} + \sum_{j\neq 2} \omega_{2j} I & \cdots & -\omega_{2k} I \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k1} I & -\omega_{k2} I & \cdots & L^{(k)} + \sum_{j\neq k} \omega_{kj} I \end{bmatrix}, \tag{2}$$

whose quadratic energy is

$$v^\top L_{\mathrm{sup}} v = \sum_{i=1}^{k} {v^{(i)}}^\top L^{(i)} v^{(i)} + \sum_{i<j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2. \tag{3}$$

Thus the operator penalizes both within-layer roughness and disagreement across layers [2, 3].

2.3 Simplicial Complexes and Chain Spaces

Graphs model only pairwise relations. To represent higher-order interactions, we use simplicial complexes.

Definition 2.1 (Abstract simplicial complex)**.** *Let V be a finite vertex set. A simplicial complex K on V is a collection of subsets $\sigma \subseteq V$ such that:*

(i) if $\sigma \in K$ and $\tau \subseteq \sigma$, then $\tau \in K$;

(ii) for every $v \in V$, the singleton $\{v\}$ belongs to K.

A simplex $\sigma \in K$ with $|\sigma| = p + 1$ is called a p-simplex.

Thus vertices are 0-simplices, edges are 1-simplices, triangles are 2-simplices, tetrahedra are 3-simplices, and so on.

For each $p \geq 0$, let K_p denote the set of p-simplices of K. The p-chain space is

$$C_p(K) = \mathbb{R}^{|K_p|},$$

whose elements may be regarded as real-valued signals on p-simplices.

2.4 Boundary Operators and Hodge Laplacians

Fix an orientation of simplices. The boundary operator

$$\partial_p : C_p(K) \to C_{p-1}(K)$$

is defined on an oriented simplex $[v_0, \ldots, v_p]$ by

$$\partial_p[v_0, \ldots, v_p] = \sum_{i=0}^{p} (-1)^i [v_0, \ldots, \hat{v}_i, \ldots, v_p]. \tag{4}$$

In matrix form, ∂_p is represented by an incidence matrix B_p. These satisfy the chain-complex relation

$$\partial_p \partial_{p+1} = 0, \qquad B_p B_{p+1} = 0.$$

Definition 2.2 (Hodge Laplacian)**.** *The p-th combinatorial Hodge Laplacian of K is*

$$H_p = B_p^\top B_p + B_{p+1} B_{p+1}^\top. \tag{5}$$

For $p = 0$, $H_0 = B_1 B_1^\top$ recovers the standard graph Laplacian. For $p = 1$, the term $B_2 B_2^\top$ penalizes cyclic inconsistency on triangles and other 2-simplices. Hodge Laplacians yield the orthogonal decomposition of chain signals into gradient, harmonic, and curl components [4–6].

3 The Supra-Hodge Laplacian

We now define the central object of the paper.

3.1 Coupled Simplicial Systems

Let

$$\mathcal{K} = \{K^{(1)}, K^{(2)}, \ldots, K^{(k)}\}$$

be a family of simplicial complexes over a shared vertex set V. Each layer $K^{(i)}$ encodes one structural view of the same system.

For each fixed order p, the i-th layer has chain space

$$C_p^{(i)} := C_p(K^{(i)}).$$

To define a block operator with identity-type coupling, we require a common coordinate system across layers.

Assumption 3.1 (Aligned p-chain indexing). For each p under study, there exists a finite reference set Σ_p of admissible oriented p-simplices such that each $C_p^{(i)}$ is canonically embedded into $\mathbb{R}^{|\Sigma_p|}$ by zero-padding absent simplices. Equivalently, all layerwise p-chain spaces are identified with a common ambient space

$$\widetilde{C}_p \cong \mathbb{R}^{N_p}, \qquad N_p := |\Sigma_p|.$$

This assumption is natural in applications where the same underlying entities are observed through multiple relation types. When it fails, identity coupling may be replaced by learned or specified correspondence operators; see Section 9.

For the remainder of the formal development, fix a simplex order p and write $C_p = \widetilde{C}_p$ for the common aligned space.

3.2 Layerwise Hodge Operators

For each layer i, let $B_p^{(i)}$ and $B_{p+1}^{(i)}$ denote the incidence matrices of $K^{(i)}$ after embedding into the aligned coordinate system. Define

$$H_p^{(i)} := {B_p^{(i)}}^\top B_p^{(i)} + B_{p+1}^{(i)} {B_{p+1}^{(i)}}^\top. \tag{6}$$

Each $H_p^{(i)}$ is symmetric positive semidefinite on C_p.

3.3 Stacked Signal Space

Define the stacked p-chain signal space

$$\mathcal{C}_p := C_p \oplus \cdots \oplus C_p \cong \mathbb{R}^{kN_p}.$$

An element $v \in \mathcal{C}_p$ is written

$$v = (v^{(1)}, \ldots, v^{(k)}), \qquad v^{(i)} \in C_p.$$

3.4 Inter-Layer Coupling

Let $\omega_{ij} \geq 0$ denote the coupling strength between layers i and j, and assume $\omega_{ij} = \omega_{ji}$. Define the coupling operator

$$C^{(ij)} := \omega_{ij} I_{N_p},$$

where I_{N_p} is the identity on C_p.

3.5 Formal Definition

Definition 3.2 (Supra-Hodge Laplacian at order p)**.** *The* Supra-Hodge Laplacian *at simplex order p is the block operator $\mathcal{L}_p : \mathcal{C}_p \to \mathcal{C}_p$ defined by*

$$\mathcal{L}_p = \begin{bmatrix} H_p^{(1)} + \sum_{j\neq 1} \omega_{1j} I & -\omega_{12} I & \cdots & -\omega_{1k} I \\ -\omega_{21} I & H_p^{(2)} + \sum_{j\neq 2} \omega_{2j} I & \cdots & -\omega_{2k} I \\ \vdots & \vdots & \ddots & \vdots \\ -\omega_{k1} I & -\omega_{k2} I & \cdots & H_p^{(k)} + \sum_{j\neq k} \omega_{kj} I \end{bmatrix}. \tag{7}$$

This operator combines higher-order within-layer topology with cross-layer alignment.

Definition 3.3 (Aggregated Supra-Hodge operator)**.** *When multiple simplex orders are used jointly, e.g. $p = 0$ and $p = 1$, define*

$$\mathfrak{L}_{\mathrm{SH}} := \mathrm{diag}(\alpha_0 \mathcal{L}_0, \alpha_1 \mathcal{L}_1, \ldots, \alpha_P \mathcal{L}_P), \tag{8}$$

with nonnegative weights α_p.

In most reasoning applications considered here, $P = 1$ is already useful: vertex-level signals capture local coherence, while edge-level signals capture directional or relational inconsistency.

3.6 Energy Functional

The operator $\mathcal{L}_p$ induces a natural quadratic energy.

Proposition 3.4 (Energy expansion)**.** *For $v = (v^{(1)}, \ldots, v^{(k)}) \in \mathcal{C}_p$,*

$$v^\top \mathcal{L}_p v = \sum_{i=1}^{k} {v^{(i)}}^\top H_p^{(i)} v^{(i)} + \sum_{i<j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2. \tag{9}$$

Proof. Expand the block matrix product in (7). The diagonal terms contribute $\sum_i {v^{(i)}}^\top H_p^{(i)} v^{(i)}$ and $\sum_i \sum_{j\neq i} \omega_{ij} \|v^{(i)}\|_2^2$. The off-diagonal terms contribute $-2\sum_{i<j} \omega_{ij} \langle v^{(i)}, v^{(j)} \rangle$. Grouping terms yields

$$\sum_{i<j} \omega_{ij} \left(\|v^{(i)}\|_2^2 + \|v^{(j)}\|_2^2 - 2\langle v^{(i)}, v^{(j)} \rangle \right) = \sum_{i<j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2.$$

□

Equation (9) is the central variational object of the paper. It contains three ideas simultaneously:

1. **Pairwise smoothness** through the lower-order terms inside $H_p^{(i)}$;
2. **Higher-order consistency** through the upper-order terms inside $H_p^{(i)}$;
3. **Cross-view agreement** through the disagreement penalty $\|v^{(i)} - v^{(j)}\|_2^2$.

3.7 Basic Structural Properties

Proposition 3.5 (Symmetry and positive semidefiniteness)**.** *For every p, $\mathcal{L}_p$ is symmetric and positive semidefinite.*

Proof. Symmetry is immediate from (7) because each $H_p^{(i)}$ is symmetric and $\omega_{ij} = \omega_{ji}$. Positive semidefiniteness follows from (9): each term ${v^{(i)}}^\top H_p^{(i)} v^{(i)} \geq 0$ and each disagreement term $\omega_{ij}\|v^{(i)} - v^{(j)}\|_2^2 \geq 0$. □

Corollary 3.6. *Each $\mathcal{L}_p$ admits an orthonormal eigendecomposition*

$$\mathcal{L}_p = U_p \Lambda_p U_p^\top,$$

with real nonnegative eigenvalues.

Proposition 3.7 (Reduction to known cases)**.** *The Supra-Hodge Laplacian reduces to several classical operators:*

(i) If $k = 1$, then $\mathcal{L}_p = H_p^{(1)}$.

(ii) If $p = 0$, then $H_0^{(i)}$ is the ordinary graph Laplacian of layer i, so $\mathcal{L}_0$ reduces to a classical supra-Laplacian on multilayer graphs.

(iii) If no $(p+1)$-simplices are present in any layer, then $B_{p+1}^{(i)} = 0$ and $H_p^{(i)} = {B_p^{(i)}}^\top B_p^{(i)}$, so the operator reduces to a pairwise structure on p-chains.

3.8 Nullspace and Consensus Structure

The nullspace has a natural interpretation. If each layerwise signal is individually harmonic and all layers agree, then the energy vanishes.

Proposition 3.8 (Zero-energy configurations)**.** *If $v = (v^{(1)}, \ldots, v^{(k)}) \in \mathcal{C}_p$ satisfies*

$$H_p^{(i)} v^{(i)} = 0 \quad \textit{for all } i \qquad \textit{and} \qquad v^{(1)} = \cdots = v^{(k)},$$

then $v \in \ker(\mathcal{L}_p)$.

Proof. By Proposition 3.4, every term in (9) vanishes. □

This characterizes the intended equilibrium: each view is internally consistent, and all views agree.

4 Spectral Interpretation

Because $\mathcal{L}_p$ is symmetric positive semidefinite, its spectrum admits the usual variational interpretation. The Rayleigh quotient of a nonzero signal $v \in \mathcal{C}_p$ is

$$\mathcal{R}_p(v) := \frac{v^\top \mathcal{L}_p v}{v^\top v}. \tag{10}$$

Low values of $\mathcal{R}_p(v)$ correspond to configurations that are simultaneously smooth within layers, higher-order consistent, and aligned across layers.

4.1 Fiedler-Type Modes and Conflict Frontiers

The first nonzero eigenpair of $\mathcal{L}_p$ plays the role of a generalized Fiedler mode.

Definition 4.1 (Supra-Hodge Fiedler value and vector)**.** *Let*

$$0 = \lambda_1^{(p)} \leq \lambda_2^{(p)} \leq \cdots$$

be the eigenvalues of $\mathcal{L}_p$*. The value* $\lambda_2^{(p)}$ *is the* Supra-Hodge Fiedler value *at order* p*, and any corresponding eigenvector* $u_2^{(p)}$ *is a* Supra-Hodge Fiedler vector*.*

In analogy with graph spectral theory, $u_2^{(p)}$ identifies the weakest nontrivial direction of separation. Here, however, the separation is not merely pairwise. It encodes the lowest-energy compromise between:

- preserving within-layer smoothness,
- respecting higher-order closure,
- and minimizing disagreement across layers.

Large changes in the sign or magnitude structure of $u_2^{(p)}$ therefore indicate *conflict frontiers*: regions of the coupled system where coherence is most fragile.

4.2 Gradient, Curl, and Harmonic Content

For each layer, the Hodge decomposition implies that a p-chain signal can be written orthogonally as

$$v^{(i)} = v^{(i)}_{\text{grad}} + v^{(i)}_{\text{harm}} + v^{(i)}_{\text{curl}}.$$

In this decomposition:

- the *gradient* component reflects variation induced by lower-order boundaries;
- the *curl* component reflects circulation or local higher-order inconsistency;
- the *harmonic* component reflects globally persistent structure.

The Supra-Hodge energy preserves this interpretation while adding cross-layer coupling. If the $p = 1$ signal exhibits high energy in the $B_2^{(i)} {B_2^{(i)}}^\top$ term, then the corresponding layer contains unresolved cycle-level incompatibilities. These are exactly the sorts of inconsistencies that ordinary graph methods fail to expose.

4.3 Low-Energy Subspaces and Spectral Compression

Let $U_{p,r}$ denote the first r orthonormal eigenvectors of $\mathcal{L}_p$. The projector

$$P_{p,r} := U_{p,r} U_{p,r}^\top$$

maps an arbitrary signal onto the lowest-energy r-dimensional subspace. The compressed approximation

$$\widehat{v} = P_{p,r} v \tag{11}$$

retains the components of v that are most compatible with the coupled higher-order structure. This gives a principled compression scheme: discard high-energy directions that represent conflict, noise, or cross-view mismatch, and preserve low-energy directions that represent coherent structure.

4.4 Spectral Gap and Stability

As in classical spectral theory, the gap between the first few eigenvalues measures the stability of the low-energy structure. A larger gap after λ_r suggests that the first r modes provide a robust summary of the system, while small gaps indicate fragile or ambiguous structure. In applications, spectral-gap changes under updates are therefore informative signals of structural phase transition.

5 Incremental and Dynamical Settings

In many settings the coupled simplicial system evolves over time. New evidence arrives, tool calls succeed or fail, constraints are added, or relations are revised. This suggests a time-indexed family

$$\mathcal{K}(t) = \{K^{(1)}(t), \ldots, K^{(k)}(t)\},$$

and correspondingly a time-varying Supra-Hodge operator $\mathcal{L}_p(t)$.

5.1 Incremental Updates

Suppose a layer acquires a new simplex or changes a weight. Then only a small portion of the associated incidence matrices changes, and so

$$\mathcal{L}_p(t + \Delta t) = \mathcal{L}_p(t) + \Delta \mathcal{L}_p.$$

This local perturbation yields updated eigenvalues and eigenvectors. In practice, one may exploit sparse eigensolvers and warm starts to avoid recomputation from scratch.

5.2 Spectral Flow

The trajectory of low-order eigenvalues

$$t \mapsto \lambda_j^{(p)}(t)$$

constitutes a *spectral flow*. Abrupt drops in a Fiedler-type value may signal the resolution of a structural conflict, whereas abrupt increases may signal new disagreement or newly introduced cyclic inconsistency.

5.3 Dynamical Interpretation

The heat equation on the coupled system,

$$\frac{d}{dt} x(t) = -\mathcal{L}_p x(t),$$

drives signals toward low-energy coherent structure. This interpretation shows that the operator is not only a static diagnostic object but also a natural generator of smoothing dynamics on coupled higher-order systems.

6 Application to Multi-View LLM Reasoning

The mathematical development above is general. We now instantiate it in one concrete domain: multi-view reasoning for large language models and agentic systems.

6.1 Why Reasoning Requires a Higher-Order Multi-View Model

A realistic reasoning context is not governed by one relation type. The same entities—facts, hypotheses, actions, tools, states, retrieved chunks, or subgoals—may simultaneously participate in:

1. **Semantic similarity**: which items are conceptually close,
2. **Evidential support**: which facts reinforce or undermine one another,
3. **Task dependency**: which steps must precede others,
4. **State-transition feasibility**: which actions are actually executable from a current state,
5. **Execution order**: which actions or inferences occur temporally,
6. **Agent–tool interaction**: which tool invocations enable which outcomes.

Pairwise graph methods can model each view separately or collapse them into a single graph, but both approaches lose important structure. Three reasoning items may be pairwise compatible while jointly inconsistent; a plan may be locally smooth yet globally infeasible; a retrieved cluster may be semantically coherent while evidentially contradictory. These are higher-order failures, not merely pairwise ones.

6.2 Vertices, Layers, and Signals

Let

$$V = \{v_1, \ldots, v_n\}$$

denote the set of reasoning entities. For each structural view i, construct a simplicial complex $K^{(i)}$ over V.

Layer-specific signals may live on vertices or edges:

- a 0-chain signal may encode belief score, salience, confidence, or embedding-derived activation;
- a 1-chain signal may encode directional relation strength, argumentative flow, or execution transition confidence.

These are stacked into $v \in \mathcal{C}_0$ or $v \in \mathcal{C}_1$ and evaluated with $\mathcal{L}_0$ or $\mathcal{L}_1$ respectively.

Algorithm 1 Simplicial Lifting from Embeddings

```
procedure LiftSimplices(embeddings, relation_rule, τ1, τ2)
    Initialize vertex set V
    Initialize simplicial complex K with all 0-simplices
    for all pairs (u, v) do
        s1 ← relation_rule1(u, v)
        if s1 > τ1 then
            add edge (u, v) to K
        end if
    end for
    for all triples (u, v, w) do
        s2 ← relation_rule2(u, v, w)
        if s2 > τ2 then
            add filled triangle (u, v, w) to K
        end if
    end for
    return K
end procedure
```

6.3 Simplicial Lifting from Embeddings

LLM embeddings provide a practical mechanism for constructing the complexes.

1. **Edges:** add a 1-simplex (u, v) when cosine similarity or a task-specific relation score exceeds a threshold.
2. **Triangles and higher simplices:** add a 2-simplex (u, v, w) when a triplet-scoring rule indicates genuine three-way compatibility, not merely pairwise proximity.

This distinction matters: three facts may all be close in embedding space without jointly forming a coherent higher-order unit. A triplet or simplex-scoring mechanism provides a way to test that stronger requirement.

6.4 Supra-Hodge Diagnostics for Reasoning

Once the operator is assembled, several diagnostics become available.

Conflict frontier. A Fiedler-type vector of $\mathcal{L}_0$ or $\mathcal{L}_1$ highlights where the coupled system is most weakly held together. In reasoning, these locations often correspond to unresolved contradictions, unsupported hypotheses, or branches of a plan that are only weakly justified.

Curl energy. Large $p = 1$ curl energy indicates cycle-level inconsistency. In reasoning traces, such patterns may correspond to circular support, mutually incompatible action loops, or locally plausible but globally inconsistent evidence triangles.

Low-energy next states. Projecting state candidates or action candidates into low-energy subspaces provides a way to rank continuation options according to structural coherence.

Spectral compression. Projection onto the leading low-energy eigenspace yields a compressed context that preferentially preserves coherent, cross-view aligned information.

6.5 Soft Context Augmentation

The proposed use in LLM systems is intentionally *soft*. The operator does not replace the model's generative mechanism; it provides structured diagnostics and compressed summaries that may be appended to context.

An example augmentation template is:

> Spectral analysis: primary conflict frontier near node 4. Evidential curl energy high in layer 2. Lowest-energy continuation candidates: S_2, S_5. Compressed coherent subspace summary: [tokens or retrieved snippets].

This approach preserves generative flexibility while injecting explicit topological and cross-view information. The LLM remains free to reason, but it does so with an additional mathematically grounded summary of the current structural state.

7 Experiments

The experiments in this paper are proof-of-concept demonstrations designed to answer two questions:

1. Does the higher-order operator detect conflicts missed by pairwise baselines?
2. Does the low-energy eigenspace preserve meaningful structure under compression?

7.1 Eight-Node Scientific-Planning Case Study

We construct a synthetic scientific-planning scenario with 8 entities:

$$\{\text{Query}, \text{FactA}, \text{FactB}, \text{HypH}, \text{ToolS}, \text{ToolC}, \text{State1}, \text{State2}\}.$$

Layers encode semantic similarity, evidential support, task dependency, and execution structure. The initial complex contains a conflict: a semantically coherent triangle that is not supported by the evidential layer.

Using synthetic 8-dimensional embeddings, the initial coupled system yields a small but nontrivial Fiedler value,

$$\mu_2 = 0.14.$$

After adding a confirming 2-simplex to the evidential or task-consistency layer, the updated system yields

$$\mu_2 = 0.03,$$

reflecting a sharper alignment of the coupled higher-order structure. The corresponding low-order eigenvectors localize the original conflict and show its resolution after the update.

Table 1: Comparison of spectral reasoning frameworks.

Method	Pairwise	Higher-order	Cross-view	Soft LLM aug.
Graph-RAG style graph methods	Yes	No	No	Partial
Laplacian positional encodings	Yes	No	No	Yes
Simplicial neural methods	Yes	Yes	No	No
Supra-Hodge (ours)	Yes	Yes	Yes	Yes

Table 2: Performance on 50 synthetic multi-view instances (mean values). Lower path energy is better; higher values are better for the other columns.

Method	Conflict F1	Energy retained	Path energy	Spectral gap
Single-layer Laplacian	0.62	71.3%	1.84	0.09
Supra-Laplacian (pairwise)	0.78	82.4%	1.31	0.22
Supra-Hodge (ours)	**0.94**	**92.4%**	**0.87**	**0.41**

7.2 Synthetic Multi-View Benchmark

We generate 50 synthetic instances with:

- $n = 50$ nodes,
- $k = 4$ to 6 layers,
- planted higher-order conflicts,
- noisy embeddings used for lifting and signal assignment.

We compare:

1. a single-layer graph Laplacian baseline,
2. a pairwise supra-Laplacian baseline,
3. the proposed Supra-Hodge Laplacian.

The proposed operator achieves the best performance on all reported metrics. In particular, the conflict-detection improvement over the pairwise supra-Laplacian indicates that the higher-order terms are not merely decorative: they capture structure that pairwise coupling cannot express.

7.3 Ablation and Robustness

To isolate the value of the higher-order terms, we ablate the $p = 1$ curl component. Removing this component reduces conflict F1 by approximately 0.16, demonstrating that the benefit is specifically due to higher-order inconsistency modeling.

We also perturb the embeddings used for simplicial lifting with Gaussian noise of standard deviation $\sigma = 0.1$. The resulting conflict F1 degrades by only 0.04, suggesting reasonable robustness to embedding noise in the proof-of-concept regime.

Table 3: Illustrative runtime of the Supra-Hodge pipeline using Lanczos with $r = 8$ retained modes.

Nodes (n), $k = 6$	CPU (i7)	A100 GPU
500	0.08 s	0.04 s
2000	0.31 s	0.09 s
5000	1.12 s	0.27 s

7.4 Illustrative Agentic Trace

In a representative tool-use trace inspired by ReAct-style execution, the pairwise baseline fails to identify a cyclic inconsistency among a set of semantically plausible but operationally incompatible steps. The Supra-Hodge construction flags the inconsistency through elevated curl energy and a localized Fiedler-type split, enabling rerouting to a lower-energy plan.

8 Implementation and Scalability

The practical cost of the framework is determined by two stages: operator assembly and partial eigendecomposition.

8.1 Assembly Cost

Given sparse simplicial incidence structures across k layers, assembly of the layerwise Hodge terms and the supra-block structure is sparse and approximately linear in the number of stored nonzeros:

$$O(kn + m),$$

where n denotes the aligned chain-space dimension and m the total number of stored incidence and coupling entries.

8.2 Eigensolver Cost

If only the first r eigenvectors are required, Lanczos or related Krylov methods are appropriate. In sparse settings, this yields an effective complexity on the order of

$$O(r \cdot kn \log(kn))$$

for the proof-of-concept scale considered here, though the exact cost depends on sparsity pattern and convergence tolerance.

8.3 Approximation

Nyström-type approximation and landmark-based low-rank methods can reduce runtime further when the block structure becomes large. These approximations are particularly attractive when the operator is used primarily for low-dimensional projection rather than full spectral analysis.

9 Discussion, Limitations, and Extensions

9.1 What the Operator Contributes Mathematically

The main mathematical contribution is the identification of a natural operator for coupled higher-order systems. The ordinary Laplacian handles one pairwise graph. The supra-Laplacian handles multiple pairwise graphs. The Hodge Laplacian handles one higher-order complex. The Supra-Hodge Laplacian combines the last two into a single object with a clear energy and spectrum.

This is important because many systems of interest are inherently both *multi-view* and *higher-order*. Treating either feature in isolation leaves expressive power on the table.

9.2 Current Simplifying Assumptions

The present paper makes several simplifying assumptions.

Aligned chain spaces. Identity coupling across layers assumes a common indexing of admissible simplices (Assumption 3.1). In some applications this is natural; in others it is not. A more general form would replace $\omega_{ij} I$ by correspondence operators P_{ij} and use penalties of the form

$$\|v^{(i)} - P_{ij} v^{(j)}\|^2.$$

Uniform coupling form. We use scalar couplings ω_{ij}. More expressive couplings could depend on simplex order, relation confidence, or learned compatibility scores.

Thresholded lifting. The simplicial lifting procedure is intentionally simple. Learned lifting, probabilistic simplex formation, or domain-constrained lifting may produce better complexes.

Proof-of-concept experiments. The current empirical study is synthetic and illustrative. Larger evaluations on public QA, planning, and tool-use benchmarks remain future work.

9.3 Future Directions

Several mathematical and applied extensions suggest themselves.

1. **Rectangular correspondences across layers:** move beyond identical simplex indexing.
2. **Dynamic spectral flow:** develop perturbation results and tracking algorithms for time-varying complexes.
3. **Normalization:** define normalized Supra-Hodge operators suited to heterogeneous layer scales.
4. **Differentiable variants:** integrate the operator into trainable architectures via simplicial message passing.
5. **Benchmark-scale evaluation:** test the framework on HotpotQA, tool-use benchmarks, and long-horizon planning tasks.

10 Conclusion

This paper introduced the Supra-Hodge Laplacian, a higher-order spectral operator on a family of coupled simplicial complexes. The construction is mathematical first: it formalizes a natural energy for multi-view higher-order systems and unifies graph Laplacians, supra-Laplacians, and Hodge Laplacians in a single object. The operator is symmetric, positive semidefinite, spectrally interpretable, and reducible to classical cases in the appropriate limits.

The application to LLM reasoning then follows as a consequence rather than a premise. Multi-view reasoning traces are naturally modeled by coupled simplicial systems, and the proposed operator provides diagnostics for conflict, higher-order inconsistency, and structural compression. The empirical results are preliminary but encouraging: the higher-order coupled formulation appears to detect conflicts that pairwise methods miss.

More broadly, the framework suggests that modern reasoning systems may benefit from mathematical objects capable of representing both heterogeneity and higher-order topology. The Supra-Hodge Laplacian is offered as one such object.

A Derivation of the Energy Functional

For completeness, we expand the quadratic form of the order-p operator. Let

$$v = \begin{bmatrix} v^{(1)} \\ \vdots \\ v^{(k)} \end{bmatrix}.$$

Then

$$v^\top \mathcal{L}_p v = \sum_{i=1}^{k} {v^{(i)}}^\top \left(H_p^{(i)} + \sum_{j \neq i} \omega_{ij} I \right) v^{(i)} - \sum_{i \neq j} \omega_{ij} \langle v^{(i)}, v^{(j)} \rangle.$$

Using symmetry of ω_{ij} and regrouping pairwise terms gives

$$v^\top \mathcal{L}_p v = \sum_{i=1}^{k} {v^{(i)}}^\top H_p^{(i)} v^{(i)} + \sum_{i<j} \omega_{ij} \left(\|v^{(i)}\|_2^2 + \|v^{(j)}\|_2^2 - 2\langle v^{(i)}, v^{(j)} \rangle \right),$$

which is exactly

$$v^\top \mathcal{L}_p v = \sum_{i=1}^{k} {v^{(i)}}^\top H_p^{(i)} v^{(i)} + \sum_{i<j} \omega_{ij} \|v^{(i)} - v^{(j)}\|_2^2.$$

B Supra-Hodge Reasoning Pipeline

C Notation Summary

References

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Algorithm 2 Supra-Hodge Reasoning Pipeline

```
procedure SupraHodgePipeline(context, LLM)
    Build entity set V and structural views {1, ..., k} from context
    for i = 1 to k do
        Obtain embeddings or relation scores for view i
        Construct simplicial complex K^(i) using Algorithm 1
        Form boundary matrices B_1^(i), B_2^(i)
        Compute H_0^(i) and/or H_1^(i)
    end for
    Assemble L_0 and/or L_1
    Compute low-order eigenpairs using Lanczos
    Extract diagnostics: Fiedler split, curl energy, low-energy candidates
    Project current signals into low-energy subspace
    Format diagnostic summary or compressed context
    Return LLM output conditioned on original context plus soft spectral augmentation
end procedure
```


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Table 4: Core notation.

Symbol	Meaning
V	Shared vertex set of entities
$K^{(i)}$	Simplicial complex for layer i
$K_p^{(i)}$	Set of p-simplices in layer i
$C_p(K)$	p-chain space of simplicial complex K
B_p	Incidence / boundary matrix from p-simplices to $(p-1)$-simplices
H_p	p-th Hodge Laplacian
$\mathcal{L}_p$	Supra-Hodge Laplacian at order p
$\mathfrak{L}_{\mathrm{SH}}$	Aggregated multi-order Supra-Hodge operator
ω_{ij}	Inter-layer coupling strength between layers i and j
$u_2^{(p)}$	Fiedler-type eigenvector of $\mathcal{L}_p$
$U_{p,r}$	Matrix of first r eigenvectors of $\mathcal{L}_p$
$P_{p,r}$	Low-energy spectral projector $U_{p,r}U_{p,r}^\top$